Rheumatoid Arthritis and Osteoarthritis

All key points for the exam (ACR guidelines) — drug table after each topic

1. RA vs OA — Overview

Feature	Rheumatoid arthritis (RA)	Osteoarthritis (OA)
Nature	**Systemic autoimmune inflammatory** disease (synovitis)	**Degenerative** joint disease (cartilage loss) — most common arthritis
Age / sex	Any age, peak 30–60 • **women 3:1**	**Older adults** (> 50) • women slightly more
Joints	**Symmetric** small joints: **MCP**, **PIP**, **wrists**, MTP • **spares DIP**	**Weight-bearing** joints: **knees**, **hips**, spine • hands: **DIP** (Heberden's nodes), PIP (Bouchard's nodes), base of thumb • often asymmetric
Morning stiffness	**> 1 hour**, improves with activity	**< 30 minutes** • "gelling" after rest
Pain pattern	Worse after rest, joint swelling, warmth	**Worse with activity**, relieved by rest • crepitus, bony enlargement
Systemic features	**Yes**: fatigue, low-grade fever, weight loss, extra-articular disease	**No**
Labs	**RF**, **anti-CCP** (most specific), **↑ ESR/CRP**, anemia of chronic disease	**Normal** ESR/CRP • negative RF
X-ray	Periarticular osteopenia, **erosions**, symmetric joint space narrowing	**Osteophytes**, asymmetric joint space narrowing, subchondral sclerosis, cysts
Treatment goal	**Stop inflammation and joint destruction → DMARDs early**	**Relieve pain, maintain function** — no disease-modifying drugs

2. Rheumatoid Arthritis: Clinical Features and Diagnosis

Clinical features

- **Symmetric polyarthritis** of small joints of hands and feet • swelling, tenderness, warmth • **morning stiffness > 1 hour**.
- **Deformities** (late): **ulnar deviation**, **swan-neck** and **boutonnière** deformities, cervical spine (C1–C2) instability.
- **Extra-articular manifestations: rheumatoid nodules**, **interstitial lung disease**, pleural effusion, **secondary Sjögren's** (dry eyes/mouth), scleritis/episcleritis, vasculitis, pericarditis, **anemia of chronic disease**, **Felty's syndrome** (RA + splenomegaly + neutropenia), **↑ cardiovascular disease** (leading cause of death), osteoporosis, ↑ lymphoma risk.

Diagnosis (2010 ACR/EULAR classification — score ≥ 6/10)

Domain	Points
Joint involvement	1 large joint (0) → 2–10 large (1) → 1–3 small (2) → 4–10 small (3) → > 10 joints including ≥ 1 small (5)
Serology	Negative RF and anti-CCP (0) • low-positive (2) • **high-positive** (3)
Acute-phase reactants	Normal CRP and ESR (0) • abnormal (1)
Duration of symptoms	< 6 weeks (0) • **≥ 6 weeks** (1)

- **Anti-CCP** is the **most specific** antibody (and predicts erosive disease) • **RF** is less specific (also in elderly, hepatitis C, Sjögren's).

Disease activity and prognosis

- Measure activity with a validated tool: **DAS28**, **CDAI**, **SDAI**, or **RAPID3** → remission, low, moderate, or high activity.

- **Treat-to-target:** goal = **remission or low disease activity** • assess every **1–3 months** during active disease • change therapy if **no improvement by 3 months** or **target not reached by 6 months**.
- **Poor prognostic factors:** high-titer RF/anti-CCP, early **erosions**, high disease activity, extra-articular disease, functional limitation.
- **Start DMARDs as early as possible** (within 3 months of diagnosis) → joint damage occurs early and is irreversible.

3. Rheumatoid Arthritis: Treatment Strategy (ACR 2021)

Situation	Recommendation
DMARD-naïve, moderate–high activity	**Methotrexate (MTX) monotherapy** = **preferred first-line** (over hydroxychloroquine, sulfasalazine, leflunomide, biologics, JAK inhibitors, and over MTX + long-term steroids)
DMARD-naïve, low activity	**Hydroxychloroquine** preferred (over other csDMARDs) • then sulfasalazine, MTX, or leflunomide
MTX inadequate dose/response	Optimize MTX: titrate to **≥ 15 mg/week** (target 20–25 mg/week) • **switch oral → SC MTX** before adding or switching drugs (better absorption, fewer GI effects)
Not at target on max-tolerated MTX	**Add a biologic (bDMARD) or a JAK inhibitor (tsDMARD)** (conditionally preferred over triple therapy) • **triple therapy** (**MTX + sulfasalazine + hydroxychloroquine**) is an alternative
Not at target on a bDMARD/tsDMARD	**Switch to a different class** (e.g., TNF inhibitor → non-TNF biologic or JAK inhibitor) • switching within TNF class is also acceptable
Sustained remission	Continue DMARDs (may **taper dose** cautiously) • **do not stop all DMARDs** (high flare risk) • taper steroids first

Key strategy rules

- **Glucocorticoids: short-term bridging** (low-dose prednisone, e.g., ≤ 10 mg/day) while waiting for DMARDs to work, or for **flares** • **avoid long-term use** (taper to the lowest dose and stop) • **intra-articular** injections for 1–2 active joints.
- **NSAIDs:** relieve **pain and stiffness only** — **do NOT slow disease progression** → never used alone in RA.
- **Never combine two biologics**, or a **biologic with a JAK inhibitor** (↑ serious infections) • biologics/JAK inhibitors **can be combined with MTX** (often more effective; reduces anti-drug antibodies).
- **Onset of action:** MTX 3–6 weeks (full 3–6 months) • HCQ 2–6 months • SSZ 1–3 months • biologics/JAK inhibitors: weeks.

Pre-treatment screening and vaccination

- **Before biologics or JAK inhibitors: TB screening** (IGRA or PPD; treat latent TB first) • **hepatitis B** (HBsAg, anti-HBc) and **hepatitis C** • CBC, LFTs, SCr.
- **Before MTX/leflunomide:** CBC, LFTs, SCr, hepatitis B/C, chest X-ray, pregnancy test.
- **Hydroxychloroquine:** baseline **eye exam**.
- **Vaccines** (ideally before starting immunosuppression): **inactivated** — influenza (yearly), **pneumococcal**, hepatitis B, HPV, COVID-19, **recombinant zoster (Shingrix)** (age ≥ 18 on immunosuppressants; especially before JAK inhibitors) • **live vaccines** (MMR, varicella, yellow fever, intranasal flu) → **avoid during biologics/JAK inhibitors**; give **≥ 4 weeks before** starting.

4. Conventional Synthetic DMARDs (csDMARDs)

Methotrexate (anchor drug)

Point	Details
Mechanism	Dihydrofolate reductase inhibitor (anti-inflammatory at low doses via adenosine)
Dose	Start **7.5–15 mg ONCE WEEKLY** (oral) → titrate to **≥ 15 mg/week within 4–6 weeks** → target **20–25 mg/week** • **SC** if inadequate response or GI intolerance • **daily dosing errors are fatal** → counsel clearly (same day each week)
Folic acid	**1 mg daily** (or 5 mg weekly, not on MTX day) → ↓ mouth ulcers, GI upset, liver toxicity, cytopenias (does not ↓ efficacy)
Adverse effects	**GI** (nausea, stomatitis), **hepatotoxicity** (fibrosis), **myelosuppression**, **pneumonitis** (dry cough, dyspnea → stop), alopecia, fatigue, infections, photosensitivity • **teratogenic** (abortifacient)
Boxed warnings	Serious toxic reactions including death (bone marrow, liver, lung, kidney, GI, skin) • **embryo-fetal toxicity**
Contraindications	**Pregnancy/breastfeeding** (stop **≥ 3 months before conception** — men and women) • **liver disease**, **heavy alcohol use** • **CrCl < 30** (renally cleared) • pre-existing blood dyscrasias, immunodeficiency
Monitoring	**CBC, LFTs, SCr** at baseline, every **2–4 weeks for 3 months**, then every 8–12 weeks • baseline **hepatitis B/C**, **chest X-ray** • limit **alcohol**
Drug interactions	**TMP-SMX** (↑ myelosuppression — avoid) • **NSAIDs** (high doses), **PPIs**, **penicillins**, probenecid (↓ renal clearance → ↑ MTX) • other hepatotoxins (leflunomide, alcohol)
Lung disease	Can be used in mild, stable lung disease (ACR 2021), but stop if pneumonitis develops

Other csDMARDs

Drug	Key points
Hydroxychloroquine (HCQ)	**Mild RA**, combination therapy • **200–400 mg/day** (**≤ 5 mg/kg/day actual body weight**) • **retinopathy** → **baseline eye exam**, then **yearly after 5 years** (sooner if risk factors) • GI upset, rash, skin hyperpigmentation, **QT prolongation**, hypoglycemia, myopathy • **no myelosuppression**, no routine labs • **safe in pregnancy** • take with food
Sulfasalazine (SSZ)	**500 mg daily–BID** → **1 g BID** (2–3 g/day) • enteric-coated with food • **GI** upset, headache, rash, **sulfa allergy**, **orange-yellow urine/skin**, stains contact lenses • **hemolysis in G6PD deficiency**, **agranulocytosis**, hepatotoxicity, **reversible oligospermia** • ↓ folate (give folic acid) • monitor CBC, LFTs • **safe in pregnancy** (with folic acid 2 mg)
Leflunomide	**20 mg daily** (10 mg if not tolerated; loading dose often skipped) • alternative to MTX • **boxed: hepatotoxicity** and **teratogenicity** • **diarrhea**, **hypertension**, alopecia, rash, weight loss, peripheral neuropathy, ILD, myelosuppression • **very long half-life** → **accelerated elimination** with **cholestyramine 8 g TID × 11 days** (before pregnancy, for toxicity) — required for **women and men** planning conception • monitor CBC, LFTs, BP • ↑ hepatotoxicity with MTX
Others (rarely)	Azathioprine, cyclosporine, minocycline, gold — limited role

5. Biologic and Targeted Synthetic DMARDs

TNF inhibitors

Drug	Dose	Notes
Etanercept (Enbrel)	**50 mg SC once weekly**	Fusion protein (soluble TNF receptor)
Adalimumab (Humira, biosimilars)	**40 mg SC every 2 weeks**	Human monoclonal antibody
Infliximab (Remicade, biosimilars)	**3 mg/kg IV at weeks 0, 2, 6**, then every 8 weeks	**Chimeric → must be given with MTX** (↓ antibodies) • infusion reactions
Certolizumab pegol (Cimzia)	400 mg SC weeks 0, 2, 4 → 200 mg q2 wk (or 400 mg q4 wk)	PEGylated, no Fc → **minimal placental transfer** (preferred in pregnancy)
Golimumab (Simponi)	50 mg SC monthly • Aria 2 mg/kg IV wk 0, 4, then q8 wk	Given with MTX

- **Boxed warnings: serious infections** (bacterial, **TB reactivation**, invasive fungal, opportunistic) • **malignancy** (lymphoma; hepatosplenic T-cell lymphoma, skin cancers).
- **Contraindications/avoid: active infection • moderate–severe heart failure (NYHA III–IV)** → worsens HF • **demyelinating disease** (MS, optic neuritis) • untreated latent TB, hepatitis B reactivation risk.
- **Other ADRs:** injection-site/**infusion reactions**, **drug-induced lupus**, **paradoxical psoriasis**, cytopenias, hepatotoxicity.
- **Counseling:** report fever/infection, hold during serious infection, no live vaccines, rotate injection sites, store in refrigerator (allow to reach room temperature).

Non-TNF biologics

Drug	Mechanism / dose	Key points
Abatacept (Orencia)	**T-cell co-stimulation blocker** (CTLA4-Ig) • IV weight-based monthly or **125 mg SC weekly**	Infections, infusion reactions, **COPD exacerbations** • lower infection risk than some biologics
Rituximab (Rituxan)	**Anti-CD20** (B cells) • **1,000 mg IV × 2 doses, 2 weeks apart**, every ~6 months	**Premedicate** (methylprednisolone, acetaminophen, antihistamine) • **boxed: infusion reactions**, **hepatitis B reactivation**, **PML**, severe mucocutaneous reactions • best in **RF/anti-CCP positive** • option with history of **lymphoma**
Tocilizumab (Actemra)	**IL-6 receptor antagonist** • 4–8 mg/kg IV q4 wk or **162 mg SC** weekly/every other week	**↑ LFTs**, **↑ lipids**, **neutropenia**, thrombocytopenia, **GI perforation** (avoid in diverticulitis) • **masks fever and CRP** during infection • boxed: serious infections
Sarilumab (Kevzara)	IL-6 receptor antagonist • **200 mg SC every 2 weeks**	Same as tocilizumab
Anakinra (Kineret)	**IL-1 receptor antagonist • 100 mg SC daily**	**Injection-site reactions** • least effective in RA (rarely used)

JAK inhibitors (tsDMARDs)

Drug	Dose	Notes
Tofacitinib (Xeljanz)	**5 mg BID** or **11 mg XR daily**	**Strong CYP3A4 inhibitors** → 5 mg daily • avoid strong inducers
Baricitinib (Olumiant)	**2 mg daily**	**Renal** adjustment • OAT3 inhibitors (probenecid) ↑
Upadacitinib (Rinvoq)	**15 mg daily**	Strong CYP3A4 inhibitors ↑ levels

- **Oral**, fast onset • can be used alone or with MTX.
- **Boxed warnings: serious infections**, **mortality**, **malignancy** (lymphoma, lung cancer), **major adverse cardiovascular events** (MI, stroke), **thrombosis** (DVT, PE, arterial) → **FDA: use only after inadequate response or intolerance to ≥ 1 TNF inhibitor** • caution in smokers, age ≥ 50 with CV

risk factors.

- **Other ADRs: herpes zoster** (give **Shingrix** first) • **↑ LDL** (lipids at 4–8 weeks) • lymphopenia, neutropenia, anemia • ↑ LFTs, ↑ CK • GI perforation.
- **Monitoring:** CBC (lymphocytes, neutrophils, Hgb), LFTs, lipids, TB screening • **avoid** with biologics or potent immunosuppressants (azathioprine, cyclosporine).

6. RA: Symptom Relief, Special Populations, and Comorbidities

Glucocorticoids

- **Bridging:** low-dose **prednisone 5–10 mg/day** for weeks–few months while DMARDs start working → **taper** and stop • **flares:** short courses • **intra-articular** triamcinolone/methylprednisolone for individual joints.
- **Long-term ADRs: osteoporosis** (give **calcium + vitamin D**; bisphosphonate if ≥ 2.5 mg/day for ≥ 3 months and at risk), hyperglycemia, hypertension, weight gain, infections, cataracts, glaucoma, adrenal suppression (taper), avascular necrosis, mood changes, skin thinning.

NSAIDs in RA

- For **pain and stiffness** only (not disease-modifying) • lowest dose, shortest time • GI, renal, CV risks (see OA section) • **separate from MTX** is usually unnecessary at RA doses but monitor.

Special populations

Population	Recommendation
Pregnancy (plan ahead)	**Contraindicated: methotrexate** (stop ≥ 3 months before), **leflunomide** (cholestyramine washout), JAK inhibitors • **Compatible: hydroxychloroquine**, **sulfasalazine** (+ folic acid), **TNF inhibitors** (**certolizumab** preferred — minimal transfer), low-dose prednisone • **avoid NSAIDs ≥ 20 weeks** (oligohydramnios; ductus closure in 3rd trimester) • RA often improves in pregnancy and flares postpartum
Heart failure (NYHA III–IV)	**Avoid TNF inhibitors** → use non-TNF biologic, JAK inhibitor, or csDMARD combination
Hepatitis B	Antiviral prophylaxis if HBsAg+ (especially rituximab) • monitor if anti-HBc+ • MTX/leflunomide caution
Previous serious infections	Prefer **csDMARDs** combination over biologics/JAK inhibitors; if needed, abatacept
Lymphoproliferative disorder history	**Rituximab** preferred
Nontuberculous mycobacterial lung disease	csDMARDs preferred; avoid biologics/JAK inhibitors if possible
CKD	Avoid MTX if CrCl < 30 • adjust baricitinib • avoid NSAIDs
Liver disease / alcohol use	Avoid MTX and leflunomide

Comorbidities to manage

- **Cardiovascular disease** (↑ risk) → control BP, lipids, smoking, diabetes • **osteoporosis** → DEXA, calcium/vitamin D • **infections** → vaccinate • **depression** • exercise, physical/occupational therapy, joint protection, weight management.

Rheumatoid arthritis drug summary

Drug	Usual dose	Key ADRs	Key DDIs / notes
Methotrexate	7.5–15 mg **once weekly** → 20–25 mg/week (PO or SC)	GI, stomatitis, **hepatotoxicity**, **myelosuppression**, **pneumonitis**, **teratogen**	**TMP-SMX**, NSAIDs, PPIs, penicillins, probenecid ↑ • alcohol • CI CrCl < 30, pregnancy
Folic acid	1 mg daily (or 5 mg weekly)	—	With MTX (and SSZ)
Hydroxychloroquine	200–400 mg/day (≤ 5 mg/kg)	**Retinopathy**, GI, **QT**, hypoglycemia, hyperpigmentation	Eye exams • other QT drugs • antacids ↓ absorption (separate 4 h)
Sulfasalazine	500 mg → 1 g BID	GI, rash, **orange urine/skin**, **G6PD hemolysis**, agranulocytosis, oligospermia	Sulfa allergy • ↓ folate, ↓ digoxin absorption
Leflunomide	20 mg daily (10 mg)	**Hepatotoxicity**, **teratogen** (boxed), diarrhea, HTN, alopecia	Cholestyramine washout • ↑ with MTX (liver) • ↑ warfarin INR
Cholestyramine (washout)	8 g TID × 11 days	Constipation, bloating	Binds other drugs — separate
Prednisone	5–10 mg/day bridging; taper	Osteoporosis, hyperglycemia, HTN, infection	Calcium + vitamin D • ↑ GI bleed with NSAIDs
Triamcinolone (intra-articular)	10–40 mg per joint (size-dependent)	Post-injection flare, local skin atrophy, transient hyperglycemia	Limit to every ~3 months per joint
Etanercept	50 mg SC weekly	**Boxed: infections (TB), malignancy** • injection site	**CI: NYHA III–IV HF, MS** • no live vaccines
Adalimumab	40 mg SC q2 wk	Same	Same
Infliximab	3 mg/kg IV wk 0, 2, 6 → q8 wk	Same + **infusion reactions**	**Must use with MTX**
Certolizumab	400 mg wk 0, 2, 4 → 200 mg q2 wk	Same	Preferred in pregnancy
Golimumab	50 mg SC monthly	Same	With MTX
Abatacept	125 mg SC weekly or IV monthly	Infections, COPD exacerbation	Screen TB, HBV
Rituximab	1,000 mg IV × 2 (2 wk apart) q6 months	**Infusion reactions**, **HBV reactivation**, PML (boxed)	Premedicate • screen HBV
Tocilizumab	162 mg SC weekly/q2 wk or 4–8 mg/kg IV q4 wk	↑ LFTs, ↑ lipids, neutropenia, **GI perforation**	Masks CRP/fever • ↓ levels of CYP3A4 substrates (restores CYP activity)
Sarilumab	200 mg SC q2 wk	Same as tocilizumab	Same
Anakinra	100 mg SC daily	Injection-site reactions, neutropenia	Renal adjustment
Tofacitinib	5 mg BID / 11 mg XR daily	**Boxed: infection, death, malignancy, MACE, thrombosis** • zoster, ↑ LDL	Strong 3A4 inhibitors → 5 mg daily • after TNF failure
Baricitinib	2 mg daily	Same	Renal adjustment • probenecid ↑
Upadacitinib	15 mg daily	Same	Strong CYP3A4 inhibitors ↑

7. Osteoarthritis: Clinical Features and Diagnosis

- **Risk factors: age** (strongest), **obesity** (**most modifiable** — especially knee), female sex, **joint injury/trauma**, repetitive occupational use, genetics, malalignment, muscle weakness.
- **Pathophysiology:** progressive **cartilage loss**, subchondral bone changes, **osteophytes**, mild synovial inflammation.
- **Symptoms: pain with activity**, relieved by rest (advanced: pain at rest/night) • **morning stiffness < 30 min** • stiffness after inactivity ("gelling") • **crepitus** • ↓ range of motion • **bony enlargement** (**Heberden's nodes** — DIP; **Bouchard's nodes** — PIP) • joint instability • **no systemic symptoms**.
- **Diagnosis:** mainly **clinical** (age ≥ 45, activity-related pain, short morning stiffness) • X-ray: **joint space narrowing**, **osteophytes**, subchondral sclerosis • labs **normal** (ESR, CRP, RF) • synovial fluid non-inflammatory (WBC < 2,000).

8. Osteoarthritis: Treatment (ACR/Arthritis Foundation 2019)

Goals

Reduce pain, maintain/improve **function** and quality of life, and **minimize drug toxicity**. **No drug changes the course of OA.**

Non-pharmacologic therapy (cornerstone for ALL patients)

Strongly recommended	Conditionally recommended
Exercise (aerobic, strengthening, aquatic) — **most important** • **weight loss** (≥ 5% of body weight for knee/hip) • **self-management programs** • **tai chi** (knee, hip) • **cane** (knee, hip — hold in the **opposite** hand) • **tibiofemoral knee brace** • **hand orthosis** (first CMC joint)	Balance training, yoga (knee), **CBT**, acupuncture, heat/cold (thermal therapy), kinesiotaping, radiofrequency ablation (knee), paraffin (hand)

Pharmacologic therapy

Recommendation	Knee	Hip	Hand
Topical NSAIDs	**Strongly recommended** (first-line)	—	Conditionally recommended
Oral NSAIDs	**Strongly recommended**	**Strongly recommended**	**Strongly recommended**
Intra-articular corticosteroids	**Strongly recommended**	**Strongly recommended** (ultrasound-guided)	Conditionally recommended
Acetaminophen	Conditional (limited benefit)	Conditional	Conditional
Duloxetine	Conditional	Conditional	Conditional
Tramadol	Conditional	Conditional	Conditional
Topical capsaicin	Conditional	—	**Conditionally against**
Chondroitin	Strongly against	Strongly against	Conditionally recommended
Intra-articular hyaluronic acid	Conditionally against	Strongly against	Conditionally against
Non-tramadol opioids	Conditionally against	Conditionally against	Conditionally against
Glucosamine	**Strongly against**	**Strongly against**	**Strongly against**

- **Also strongly against (all joints):** bisphosphonates, hydroxychloroquine, methotrexate, TNF inhibitors, IL-1 inhibitors, platelet-rich plasma, stem cell injections, fish oil, vitamin D (for OA).

Topical NSAIDs

- **Diclofenac 1% gel**: **2 g** (upper extremity) or **4 g** (lower extremity) **QID** • max **32 g/day** total (≤ 8 g/day per upper joint, ≤ 16 g/day per lower joint) • also diclofenac 1.5% solution, patch.
- **Preferred** for **knee** (and hand) OA, especially in **elderly (≥ 75)** or with **GI, renal, or CV risk** → much **lower systemic exposure**.
- ADRs: **application-site reactions** (dryness, rash) • counsel: apply to **intact skin**, **wash hands** after, avoid eyes, no heat or occlusive dressings, avoid sun exposure on treated area, don't shower/bathe for 1 hour • avoid combining with oral NSAIDs.

Oral NSAIDs

- **Most effective oral drugs** for OA pain • use the **lowest effective dose** for the **shortest time**, consider PRN use.
- **Boxed warnings: cardiovascular thrombotic events** (MI, stroke; contraindicated after CABG) and **GI bleeding, ulceration, perforation**.
- **GI risk** (age > 65, prior ulcer/bleed, anticoagulants, steroids, SSRIs) → add **PPI** (or misoprostol) or use **celecoxib** (COX-2 selective) • highest GI risk → celecoxib + PPI.
- **CV risk: naproxen** has the lowest CV risk • avoid NSAIDs in high CV risk if possible.
- **Renal:** ↓ GFR, sodium/water retention, edema, **hyperkalemia**, ↑ BP → **avoid in CKD (CrCl < 30)**, **heart failure**, cirrhosis.
- **Elderly (Beers):** avoid chronic use unless alternatives fail and a PPI is added; avoid indomethacin and ketorolac.
- **Pregnancy:** avoid at **≥ 20 weeks** (oligohydramnios, fetal renal problems) and in 3rd trimester (premature ductus closure).
- **Aspirin interaction: ibuprofen** (and naproxen) can block aspirin's antiplatelet effect → take **immediate-release aspirin ≥ 30 minutes before** ibuprofen, or ibuprofen ≥ 8 hours before aspirin • celecoxib and diclofenac do not interfere.

NSAID DDI	Effect
Anticoagulants, antiplatelets, SSRIs/SNRIs, corticosteroids	↑ **GI bleeding**
ACE inhibitor/ARB + diuretic ("triple whammy")	↑ **acute kidney injury**
Antihypertensives, diuretics	↓ antihypertensive/diuretic effect
Lithium	↑ lithium levels (toxicity)
Methotrexate (high dose)	↑ MTX toxicity
Potassium-sparing drugs	↑ hyperkalemia
Celecoxib	**Sulfonamide allergy** (caution) • **CYP2C9** substrate (fluconazole ↑ levels)

Other drugs

Drug	Key points
Acetaminophen	Limited efficacy but safe → for mild pain or when NSAIDs contraindicated • max **3 g/day** (up to 4 g) • **≤ 2 g/day** with liver disease or regular alcohol use • watch **combination products** (hidden acetaminophen) • regular use ↑ **INR on warfarin**
Topical capsaicin (0.025–0.1%)	Depletes substance P • **knee** OA (not hand) • apply 3–4 times daily • **burning** at first; takes **2–4 weeks** for full effect • wash hands, avoid eyes and broken skin
Duloxetine	30 mg → 60 mg daily • for chronic OA pain, especially with **widespread pain or depression** • nausea, dizziness, hepatotoxicity, serotonin syndrome risk
Tramadol	When other options fail or are contraindicated • 50–100 mg q4–6h (max 400 mg/day; 200 mg if CrCl < 30; 300 mg if elderly > 75) • **seizures**, **serotonin syndrome** (SSRIs, SNRIs, MAOIs), hypoglycemia, hyponatremia, dependence • **CYP2D6** (poor metabolizers ↓ effect)
Other opioids	Conditionally **against** • only if all alternatives are exhausted
Intra-articular corticosteroids	**Triamcinolone** 40 mg (or ER triamcinolone 32 mg for knee), methylprednisolone • relief for **weeks** • limit to **every 3 months** (≤ 3–4/year per joint) • post-injection flare, transient **hyperglycemia** in diabetes, infection (rare), possible cartilage loss with repeated use
Intra-articular hyaluronic acid	Not recommended (minimal benefit)
Glucosamine	**Not recommended** (no benefit) • shellfish source; may ↑ INR with warfarin, ↑ glucose
Chondroitin	Not recommended for knee/hip • may help **hand** OA

Surgery

Total joint replacement (knee, hip) for **severe** pain and functional loss not controlled by other treatments • osteotomy in younger patients.

Choosing therapy (practical)

Patient	Preferred approach
Knee OA, any patient	Exercise + weight loss → **topical NSAID** → oral NSAID → intra-articular steroid
Elderly (≥ 75) or multiple comorbidities	**Topical NSAID** first • acetaminophen • avoid chronic oral NSAIDs
High GI risk	Topical NSAID • **celecoxib + PPI** if oral needed
CKD or heart failure	Topical NSAID, acetaminophen, duloxetine, intra-articular steroid • **avoid oral NSAIDs**
High CV risk	Avoid NSAIDs if possible • if needed: **naproxen** (+ PPI if on aspirin)
On low-dose aspirin	Avoid ibuprofen timing interaction • add PPI
Widespread pain or depression	**Duloxetine**
Hip OA	Oral NSAIDs, ultrasound-guided intra-articular steroid (topical NSAIDs less useful for deep joints)
Hand OA	Oral or topical NSAIDs, orthosis, chondroitin, intra-articular steroid

Osteoarthritis drug summary

Drug	Usual dose	Key ADRs	Key DDIs / notes
Diclofenac 1% gel	2 g (upper) or 4 g (lower) QID • max 32 g/day	Application-site reactions	Wash hands • avoid with oral NSAIDs • first-line knee/hand
Capsaicin cream	0.025–0.1% TID–QID	**Burning**, erythema	2–4 weeks for effect • avoid eyes • knee only
Acetaminophen	500–1,000 mg q6h (max 3 g/day; ≤ 2 g with liver disease/alcohol)	Hepatotoxicity	Hidden in combinations • ↑ INR with warfarin
Ibuprofen	400–800 mg TID–QID (max 3,200 mg/day; OTC 1,200)	GI bleed, renal, HTN, CV	**Blocks aspirin's antiplatelet effect** • lithium, MTX, ACEi/ARB + diuretic
Naproxen	250–500 mg BID (max 1,500 mg/day)	GI bleed, renal	**Lowest CV risk** • same DDIs
Celecoxib	100 mg BID or 200 mg daily	CV (boxed), renal, less GI	**Sulfa allergy** caution • CYP2C9 (fluconazole ↑) • no aspirin interaction
Meloxicam	7.5–15 mg daily	GI, renal, CV	Some COX-2 preference at low dose
Diclofenac (oral)	50 mg BID–TID (max 150 mg/day)	**Hepatotoxicity**, GI, CV (higher)	Check LFTs
Diclofenac/misoprostol	50/200 mcg BID–TID	Diarrhea	**CI in pregnancy** (boxed)
Omeprazole (GI protection)	20 mg daily	Long-term: C. diff, fractures, ↓ Mg, ↓ B12	CYP2C19 (↓ clopidogrel)
Misoprostol	200 mcg QID	**Diarrhea**, cramps	**CI in pregnancy**
Duloxetine	30 → 60 mg daily	Nausea, dizziness, hepatotoxicity	Serotonin syndrome • avoid CrCl < 30, heavy alcohol
Tramadol	50–100 mg q4–6h (max 400 mg; 200 mg if CrCl < 30)	**Seizures**, **serotonin syndrome**, dependence, hypoglycemia	SSRIs/SNRIs/MAOIs • CYP2D6
Triamcinolone (intra-articular)	40 mg (knee) • ER 32 mg (knee)	Post-injection flare, hyperglycemia	≤ every 3 months
Hyaluronic acid (intra-articular)	Product-specific series	Local pain, swelling	Not recommended
Glucosamine / chondroitin	Glucosamine 1,500 mg/day • chondroitin 800–1,200 mg/day	GI upset	Glucosamine not recommended; may ↑ INR

Exam Traps and Frequently Tested Points

Rheumatoid arthritis

1. **RA:** symmetric small joints (MCP, PIP, wrists), **spares DIP**, morning stiffness **> 1 hour**, systemic features.
2. **Anti-CCP** = most specific antibody; RF less specific.
3. **Start DMARDs early** (within 3 months) • treat-to-target (remission/low activity) • reassess at 3 and 6 months.
4. **Methotrexate** = first-line (moderate–high activity) • **hydroxychloroquine** for low activity.
5. **MTX is ONCE WEEKLY** • target 20–25 mg/week • switch to **SC** before adding drugs • **folic acid 1 mg daily**.
6. **MTX:** hepatotoxicity, myelosuppression, pneumonitis, stomatitis • **CI: pregnancy** (stop 3 months before), liver disease, alcohol, CrCl < 30.
7. **MTX + TMP-SMX** → severe myelosuppression • NSAIDs, PPIs, penicillins ↑ MTX.

8. **Hydroxychloroquine:** retinopathy (eye exam baseline, yearly after 5 years), ≤ 5 mg/kg/day, safe in pregnancy, no lab monitoring.
9. **Sulfasalazine:** sulfa allergy, orange urine, G6PD hemolysis, agranulocytosis, oligospermia.
10. **Leflunomide:** hepatotoxic and teratogenic → **cholestyramine washout** (men and women).
11. **Triple therapy** = MTX + SSZ + HCQ.
12. **Not at target on MTX** → add biologic or JAK inhibitor.
13. **Steroids** = bridge only (low dose, taper) • **NSAIDs** = symptoms only, not disease-modifying.
14. **Never combine two biologics or a biologic + JAK inhibitor**.
15. **Before biologics/JAK: TB and hepatitis B** screening • no live vaccines • give Shingrix and inactivated vaccines first.
16. **TNF inhibitors CI: NYHA III–IV heart failure**, **MS/demyelinating disease**, active infection.
17. **Infliximab** → must be given with MTX (chimeric) • **certolizumab** → preferred in pregnancy.
18. **Rituximab** → infusion reactions, HBV reactivation, PML; premedicate • preferred with lymphoma history.
19. **Tocilizumab** → ↑ LFTs, ↑ lipids, **GI perforation**, masks CRP/fever.
20. **JAK inhibitors** → boxed: infection, death, cancer, MACE, thrombosis • after **TNF failure** • herpes zoster • ↑ LDL • tofacitinib + strong CYP3A4 inhibitor → 5 mg daily.
21. **Pregnancy:** avoid MTX, leflunomide, JAK inhibitors • OK: HCQ, SSZ, TNF inhibitors (certolizumab), low-dose prednisone.

Osteoarthritis

1. **OA:** weight-bearing joints, DIP (Heberden's) and PIP (Bouchard's), stiffness **< 30 min**, pain with activity, normal labs.
2. **Obesity** = most modifiable risk factor • **exercise and weight loss** = cornerstone.
3. **Cane in the opposite hand** of the affected knee/hip.
4. **Topical NSAIDs** → first-line for **knee** OA, especially elderly (≥ 75) and comorbidities • max 32 g/day diclofenac gel.
5. **Oral NSAIDs** = most effective oral drugs • lowest dose, shortest time.
6. **GI risk** → PPI or celecoxib (+ PPI if highest risk) • **CV risk** → naproxen lowest • avoid NSAIDs in CKD, HF, ≥ 20 weeks pregnancy.
7. **Ibuprofen blocks aspirin** → aspirin 30 min before or ibuprofen 8 h before.
8. **NSAID DDIs:** anticoagulants/SSRIs (bleeding), ACEi/ARB + diuretic (AKI), lithium, methotrexate.
9. **Acetaminophen** → limited benefit; max 3 g/day (2 g with liver disease/alcohol).
10. **Capsaicin** → knee (not hand); burning; 2–4 weeks.
11. **Duloxetine** → OA with widespread pain/depression • **tramadol** → if others fail (seizures, serotonin syndrome).
12. **Intra-articular steroids** → strongly recommended (knee, hip) • max every 3 months • hyperglycemia in diabetes.
13. **Not recommended: glucosamine**, chondroitin (knee/hip), hyaluronic acid, non-tramadol opioids, PRP, stem cells, MTX, HCQ, biologics.
14. **No drug slows OA progression** • severe disease → joint replacement.

> **Note:** Doses and recommendations differ slightly between references (ACR 2021 RA, ACR/AF 2019 OA, EULAR). **Use the numbers in the reference you are studying for the exam.**

Good luck on your exam!